

FunAdemy

حيث تلتقي المتعة بالأكاديمية

WEBSITE DESIGN — FUNADEMY.IO

تصميم الموقع الكامل — الهيكل والصفحات والمكونات والنصوص والمواصفات التقنية

STEM concept · sitemap · page specifications · component library · bilingual copy deck · conversion flows · backend, data model and API · admin · deployment · launch QA · the coded site (funademy-web)

Version 6.0 · September 2026 · Internal — Founder review · STEM-first design in the brand's navy and yellow, coded front to back

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1 · What This Site Has to Do

ما الذي يجب أن يفعله الموقع

funademy.io has one job: turn a Saudi parent's first visit into a booked free trial class. Everything else — the two tracks, the pricing, the campaign, the schools page — exists to make that decision easy in under three minutes on a phone. The site is Arabic-first (RTL default, English at /en), speaks in the brand voice of a confident engineer-father, and shows the brand's one visual idea — the kite that is also a graduation cap — as the first thing on screen.

لموقع funademy.io مهمة واحدة: أن يحوّل الزيارة الأولى لولي أمر سعودي إلى حجز حصة تجريبية مجانية. كل ما عداه — المساران، الأسعار، الحملة، صفحة المدارس — موجود ليجعل هذا القرار سهلًا في أقل من ثلاث دقائق على الجوال. الموقع عربي أولًا (اتجاه من اليمين افتراضيًا، والإنجليزية على /en)، وصوته صوت أب مهندس واثق، ويعرض فكرة العلامة البصرية الواحدة — الطيارة التي هي قبعة تخرج — كأول ما تراه العين.

Delivered with this document

المسلَّم مع هذه الوثيقة

- index.html — a working single-file prototype of the full site: eight pages, Arabic/English toggle, the age rail, the live ticker, the booking form, all brand tokens. Opens on any phone or laptop; hands to the developer as the visual and copy reference.
- This document — the sitemap, the page-by-page specification, the component library, the copy deck, conversion flows, technical and SEO specification, and the launch QA gate.
- Screens — desktop and mobile captures of the prototype at the end of the document.

The five decisions behind the design

القرارات الخمسة وراء التصميم

Decision	Choice	Why
One memorable element	The hero kite flips once into the graduation cap on load, then sways ±4°	The brand promise as an optical illusion — fun and academy are the same shape; everything else on the page stays quiet
The useful interactive	An age rail (4 → 18) that shows the engineering program and the SkyLab altitude for that age	Parents think in ages, not program names; two tracks become one answer
Two tracks, one brand	Engineering and robotics (seven programs, small curriculum credit) and SkyLab (original) presented as equals under FunAdemy	FunAdemy leads; the licensed mark appears only in the fixed footer credit line
Night and sand rhythm	Dark blueprint-grid sections alternate with paper-sand sections; the sand stripe is the only divider	The brand's 60/30/10 rule on a long scrolling page without visual fatigue
Conversion everywhere	A gold 'book the free class' button in the header, the hero, the tracks, the pricing, and every page head; WhatsApp as the second door	The trial class is the product's proof; nothing should be more than one tap from it

2 · A STEM Academy, Not an S·E·M One — the Third Design

أكاديمية STEM كاملة — التصميم الثالث

The third design starts from the correction that mattered: FunAdemy is a STEM academy, and the site has to show all four pillars — science, technology, engineering and math — as one thing a child grows through, not as a flight metaphor that quietly drops the T. So the first thing on the page is not a picture. It is four working laboratories the parent can use with a thumb: a density experiment (science), a block-coded kite (technology), a gear train (engineering) and a shadow clock (math). Each one is a real moment from a real FunAdemy program, and each one answers a touch with a number.

التصميم الثالث ينطلق من التصحيح المهم: FunAdemy أكاديمية STEM، وعلى الموقع أن يُظهر الأركان الأربعة — العلوم والتقنية والهندسة والرياضيات — كشيء واحد ينمو فيه الطفل، لا كاستعارة طيران تُسقط التقنية بهدوء. لذلك أول ما تراه على الصفحة ليس صورة، بل أربعة مختبرات تعمل بإبهام ولي الأمر: تجربة كثافة (علوم)، وطيارة تُبرمج بالأوامر (تقنية)، وقطار تروس (هندسة)، وساعة ظل (رياضيات). كل واحد منها لحظة حقيقية من برنامج حقيقي في FunAdemy، وكل واحد يجيب اللمسة برقم.

Element	What it is	The STEM it shows
STEM playground (hero)	Four live micro-labs: Sink or float? (salt slider, egg rises at ρ ≥ 1.03) · Code the kite (forward / turn blocks, RUN, reach the star) · Gear ratio (16 : 8, the small gear turns twice, opposite way) · Shadow clock (sun angle → shadow = 50 ÷ tan θ)	S · T · E · M, one each, in the first screen
STEM map by age	A transit map: four lines S, T, E, M across ages 4–18; every program is a station on every line it covers; an age cursor lights the stations for that child and deals the matching STEM cards	Coverage of all four pillars at every age, at a glance
STEM cards	Paper cards with an S·T·E·M stamp row (lit letters = pillars covered), ages, minutes, seats, format, an icon and a book button	Each program declares its pillars
Four pillars	S · T · E · M panels with the programs, three promises each, and pillar textures (lab bubbles, circuit dots, blueprint gold, graph paper)	The academy's structure in one row
Live labs board	This week's live classes with a STEM column (T·E, S·M …) fed by the admin schedule	Real times, real seats, and the pillars of each lab
The FunAdemy loop	Build · test · improve · present — the shape of every session	The engineering method as the teaching method
One price ticket, T-minus, national route	The fare ticket with the free first class as a tear-off stub that books; the countdown to SkyLab Mission 1; the campaign route to Mini Watan	Conversion, urgency and the national story
The mark	The header kite flips into the graduation cap under the mouse and on keyboard focus	The brand's promise, everywhere

Design language

لغة التصميم

- One sky, four textures: the page stays FunAdemy night; each pillar carries its own surface — lab bubbles for S, circuit dots for T, gold blueprint for E, graph paper for M — and its own letter colour (cyan, white, gold, sand) used consistently on lines, stamps and panels.

- Baloo Bhaijaan 2 for headlines in both languages and for the giant pillar letters; IBM Plex Sans Arabic for reading; IBM Plex Mono only for instrument data (board rows, map labels, lab readouts).
- Every interaction returns a number: salt → density, blocks → a path, angle → degrees, sun → centimetres. Nothing moves by itself except the board's arrival and the countdown.
- Paper objects (cards, ticket, FAQ) are the only light surfaces; gold is action; Homeland Green appears once.

The palette: back to navy and yellow

اللوحة اللونية: العودة إلى الكحلي والأصفر

Version 6 returns the site to the brand's own pair. Blueprint Night is the ground of every page — a quiet navy gradient with a faint grid, no white sections and no purple — and Star Gold is the one colour that competes with it: every action, the second line of the hero, the section numbers, the E pillar, the fare stub, the seats on the board, the kite. White is for words and the paper objects (STEM cards, the ticket, the FAQ); the brand's cyan and sand are held back for the S and M pillars and thin lines. The result is the contrast the brand was built on — night and gold — without the brightness of a white canvas.

يعيد الإصدار السادس الموقع إلى زوج العلامة نفسه. الكحلي الليلي أرضية كل صفحة — تدرج كحلي هادئ بشبكة خفيفة، بلا أقسام بيضاء وبلا بنفسجي — والذهبي هو اللون الوحيد الذي ينافسه: كل فعل، والسطر الثاني من العنوان، وأرقام الأقسام، وركن الهندسة، وقسيمة التذكرة، والمقاعد على اللوحة، والطيارة. الأبيض للكلمات وللأشياء الورقية (بطاقات STEM، التذكرة، الأسئلة)، والسماوي والرملي محجوزان لركني العلوم والرياضيات والخطوط الرفيعة. النتيجة هي التباين الذي بُنيت عليه العلامة — الليل والذهب — من دون سطوع الخلفية البيضاء.

Colour	Values	Jobs on the site
Navy	#0B2263 → #091D52 → #051036 · panels #03102E · Navy Ink #101A38	Page ground, panels (playground, board, map), footer; ink on paper objects
Yellow	#FFCA17 · shadow #D9A600	Every button, the hero's second line, section numbers, the E pillar and E line, the kite, seats and readouts, the free-class stub, the sand→gold stripe
White	#FFFFFF · Paper Sand #F7EFDD	Text; STEM cards, the price ticket and the FAQ as paper objects; the T pillar
Cyan · Sand (accents)	#22DAFF · #F2CD7E / #F7E3B2	The S and M pillars and lines, live dots, the campaign route — thin, never surfaces

The kite flies

الطيارة تطير

The hero's kite no longer sits still: it glides along a gentle curve across the top of the page on a CSS motion path (22 seconds each way, eased), sways as it flies, leaves a dashed gold trail, and flips into the graduation cap when the mouse touches it. It stops under 'reduce motion'.

Classroom pictures

صور الفصول

The design carries a four-picture band — building, coding, experimenting, presenting — in the photo slots the brand needs. No photo library or image generator is reachable from this workspace, so the slots ship with original flat-vector scenes drawn in the brand's illustration style (thick Navy Ink outlines, rounded joints, gold and cyan fills) at 1600 × 1200. They are placeholders: the brand rule stands — real class photos with written parent consent within the first month; until then, AI images may be generated from the prompt pack below at 1600 × 1200 or larger and dropped into /public/scenes.

Slot	Photo brief	Brand-locked prompt (Midjourney / DALL·E)
build.svg → build.jpg	A Saudi boy or girl, 7–10, at a classroom table building a motorised gear model; hands and the model are the hero; warm natural light; navy-and-gold room	Documentary photo, Saudi child age 8 building a small motorised gear model at a bright classroom table, hands in focus, joyful concentration, navy and gold interior details, warm natural light, 35mm, no logos, no text --ar 4:3
code.svg → code.jpg	A child at a laptop programming a small robot beside it; screen glow on the face; no readable code brands	Documentary photo, Saudi child age 10 at a laptop coding a small toy robot on the desk, screen glow, focused smile, navy classroom with gold accents, warm light, 35mm, no logos, no readable text --ar 4:3
experiment.svg → experiment.jpg	A child at a kitchen table with a measuring cylinder and a beaker from the home kit, on a video call; a parent's hands nearby	Documentary photo, Saudi child age 11 doing a kitchen science experiment with a graduated cylinder and beaker at home during a video class, tablet in frame, parent hands nearby, warm home light, 35mm, no logos, no text --ar 4:3
measure.svg → measure.jpg	A teenager presenting a chart on a board to family, pointing at a graph; Launch Day mood	Documentary photo, Saudi teenager age 15 presenting a hand-drawn graph on a poster board to family in a living room, proud, gold and navy poster colours, warm evening light, 35mm, no logos, no readable text --ar 4:3

Removed

ما أُزيل

- The Home SkyLab Kit strip (the five box samples) — the kits remain in the SkyLab pages' copy and in the kit design document, not on the home page.
- The white canvas and the purple accent of v4–v5.

What improved in v5

ما تحسّن في الإصدار الخامس

- Hero: a faint kite watermark anchors the page; four trust badges (licensed, certificate per level, groups of 8–16, parent report) sit under the buttons; the playground has a title and a LIVE DEMO pulse, and every lab links to the program it comes from.
- Proof band: the academy in five numbers — ages 4–18, 12 programs and levels, 50 SkyLab missions, groups of 8–16, one free class.
- STEM map: a legend, and a dashed gold path that draws the child's route from the chosen age to 18 on both tracks.
- Four pillars: an icon per pillar and a 'try it' link back to the matching lab.
- New section — the five Home SkyLab Kits with the real box designs from the kit specification.
- The FunAdemy loop as a ring diagram with the kite in the centre, beside four step cards.
- SkyLab page: the fifty Year A missions, ten per altitude, in the child's language.
- Program pages: the session as a minute timeline (build · test · improve · present) and a 'who it is for' row; the booking page shows 'what happens next' beside the form.
- Every section title carries the marine-to-purple-to-yellow stripe; the footer has the lockup, a full navigation row and the credit lines; Kits joins the header navigation.

3 · Design System (tokens)

نظام التصميم

Tokens

القيم الأساسية

Token	Value	Use
Blueprint Night → Deepest Navy	#091D52 → #051036 gradient + 40 px grid at 5.5% white	Dark sections, header, footer
Paper Sand	#F7EFDD	Light sections and cards
Star Gold / Gold Shadow	#FFCA17 / #D9A600	Every action button (3D shadow 5 px), the kite, price numerals
Navy Ink	#101A38	All text on light backgrounds — never pure black
Cyan Glow	#22DAFF	Live ticker dots, dashed annotations, focus rings — never surfaces
Sand ladder	#F7E3B2 → #F2CD7E → #E5AE4A	The stripe divider, the SkyLab altitude rungs, the campaign phases
Homeland Green	#0E6B3A	Only the 'our nation' phase and Mini Watan
Display type	Baloo Bhaijaan 2, 800 (700 for buttons)	Headlines in Arabic and Latin, the wordmark, prices
Body type	IBM Plex Sans Arabic 400–700	Everything else; 16 px base, 1.7 line height
Radii	24 px cards · 16 px tiles · pill buttons	One radius per hierarchy level, not one radius everywhere
Grid	Max content 1180 px; 20 px gutters; 72 px section padding	Two columns from 760 px, four from 960 px

Motion

الحركة

- One orchestrated moment: the hero flip (2.6 s) then the kite sway. Nothing else moves on load.
- Motion that answers a tap: the gold button presses down 4 px; the shine sweeps on hover; the FAQ opens; the booking form confirms in place.
- The live ticker scrolls; it stops entirely under prefers-reduced-motion, as does the flip.

Brand rules enforced in code

قواعد العلامة المطبّقة في الكود

- The name is 'FunAdemy' in Latin script inside Arabic sentences — no transliteration anywhere.
- Geography: 'الرياض / Riyadh' only; no district or neighbourhood names in copy, testimonials, schedules or metadata.
- The credit line is fixed text in the footer: FunAdemy · e² Young Engineers · Falcon Eye Ltd — License 4030388340. The licensed mark appears nowhere else; SkyLab is labelled an original FunAdemy program.
- No multicolour stripes; the sand stripe is the only decorative device. No pure black. Gold only for actions and the mark.

- Testimonials are real people or clearly marked placeholders — the prototype marks its three as placeholders.

4 · Sitemap and Navigation

خريطة الموقع والتنقل

Route (AR / EN)	Page	Job	Primary action
/ · /en	Home	Convince and route: hero, live ticker, age rail, two tracks, how it works, price, campaign, testimonials, FAQ	Book the free class
/programs	Engineering & robotics	Seven programs in three fields with the learning path	Book the free class
/programs/[slug]	Program detail (spec §4.3)	One program: field, ages, length, cap, format, what they build, schedule slots	Book this program
/skylab	SkyLab	The original math & science program: mission arc, five altitudes, kit, Flight Log, pilot	Reserve a seat in Mission 1
/pricing	Pricing	One price, what it includes, payment, family example	Start with the free class
/book	Book the free trial	The form and the WhatsApp alternative	Send the request
/campaign	National campaign	Our children build our nation: four phases, Mini Watan	Register your school
/schools	For schools and partners	Free Engineering Day, club, curricular program, fairs	Request a school proposal
/contact	Contact	WhatsApp, email, social; Riyadh in person, online for all	WhatsApp
/privacy · /terms	Legal	PDPL notice, data residency, terms of enrolment	—
/admin	Admin (spec §7)	Slots, bookings, testimonials, FAQ, counters	—

Header and footer

الترويسة والتذييل

- Header (sticky, navy glass): kite + wordmark; Programs · SkyLab · Pricing · National campaign · For schools · Contact; language pill (EN / عربي); the gold 'Book a free class' button. Below 960 px the menu collapses into a drawer with the same items and the button first.
- Footer: four columns (about, programs, links, contact) and the credit block — the fixed credit line, the SkyLab originality line, © Falcon Eye Ltd · Riyadh · funademy.io, privacy.

5 · Page Specifications

مواصفات الصفحات

4.1 Home

الرئيسية

Section	Content and copy (AR first)	Component
Hero	لا نعلّم أطفالنا عن الهندسة — بل نصنع منهم مهندسين. / We don't teach kids about engineering — we make them engineers. Whisper line: live classes · mechanics, robotics, coding · math and science · from the first brick to the first line of code. Buttons: احجز الحصة التجريبية المجانية (gold) · واتساب (ghost). Chips: ages 4–18 · Riyadh, online for everyone · global curriculum, certificates · small groups 8–16.	Kite-Cap flip, dashed annotations (kite = fun, cap = academy), orbit ring
Live ticker	17:00 الأحد · مباشر — Smartivo [3 مقاعد] … eight slots from the admin schedule; seats left in gold	Ticker (fed by /admin slots)
Age rail	4 … 18 كم عمر طفلك؟ → the engineering program(s) for that age and the SkyLab altitude, each with ages, minutes, cap, format and one line	Age rail + program tiles
Two tracks	Engineering & robotics: three fields, seven programs, small curriculum credit; SkyLab: one Saudi mission a month, four live sessions, the home kit, the altitude ladder	Glass cards, altitude ladder
How it starts	Free trial → choose the program together → the kit arrives → weekly live sessions with a parent report → Launch Day and certificate	Five numbered steps (a real sequence)
Price	1,380 ريال / طفل / برنامج شهريًا; six value lines; first class free; 'try before you decide' glass card	Price block + CTA card
Campaign band	أطفالنا يبنون وطننا: بيتي → حيّي → مدينتي → وطننا; Mini Watan on National Day 2027	Phase strip (sand ladder + green)
Parents say	Three quotes — real names and real words only; the prototype shows marked placeholders	Glass quote cards
FAQ	Six questions: where classes happen (engineering and coding in person in Riyadh; SkyLab online); what we need at home; certificates; group size; mid-year joining; payment	Accordion

4.2 Programs

البرامج

Page head: title, the learning path in one line (Big Builders → Smartivo → Bricks Challenge → AlgoPlay → Galileo Technic → RoboToys → AlgoC). Seven STEM cards, each with the field tag, the pillar stamps, the name, ages · minutes · cap · format, and the Arabic one-liner from the official catalog. Every engineering and coding program is in person in Riyadh; only SkyLab is online. Under the grid: the curriculum credit line. One gold button.

4.3 Program detail (to build; not in the prototype)

صفحة البرنامج

Block	Spec
Head	Field tag, program name, ages, session length, class cap, format; gold 'Book this program' button; the age rail's other track suggested ('Also for this age: SkyLab Glide')

Block	Spec
What they build	Three to five model or project names in plain words, one sentence each; one photo per program (real class photo within the first month; AI placeholder allowed at launch with the navy overlay)
A session	The session rhythm in four lines (build · test · improve · present) — never the licensed lesson plans
Schedule	Slots for this program from /admin with seats left — every engineering and coding program is attended in person at the FunAdemy classroom in Riyadh
Certificate	Level certificate line: 'FunAdemy · curriculum by e² Young Engineers' (Lockup B, small)
FAQ	Three questions specific to the program; link to the full FAQ

4.4 SkyLab

مختبر السماء

Head: 'FunAdemy SkyLab — مختبر السماء' with three mission questions as the lede (the Asr shadow, the pilgrim's litres, the satellite's speed). Body: the four-session arc (التساؤل · التحقيق · الرياضيات · يوم الإطلاق), the five-altitude table (level, ages, minutes, what they learn), three cards (Home SkyLab Kit, Flight Log and badges, Arabic first), and the pilot block: Lift-Off and Glide from October 2026, all levels from January 2027, seats limited — 'Reserve your child's seat in Mission 1'.

4.5 Pricing

الأسعار

The number first: 1,380 SAR / child / program a month · first class free. A key-value card of what the subscription includes (sessions, length, group size, parent report, certificate, Home SkyLab Kit included with a three-month commitment, sibling −10%), a payment card (Mada, cards, STC Pay, Tamara, e-invoice, pause at month end) and a family example (two children: 2,622 SAR). No other prices anywhere on the site — school, daycare, mall and camp pricing stay in proposals.

4.6 Book the free trial

حجز الحصة التجريبية

Parent's name · mobile · child's age (4–18) · program (the seven programs and five SkyLab altitudes) · preferred time (three evening windows and Saturday morning in person) · city (Riyadh prefilled) · notes. Submit shows an in-place confirmation: request received, we call within 24 hours. A WhatsApp button beside submit. A one-line PDPL note under the form. The record is saved to the platform database and triggers a WhatsApp/email notification to admin.

4.7 Campaign, Schools, Contact

الحملة والمدارس والتواصل

- Campaign: the four phases as a strip with term labels; the Mini Watan card (23 September 2027, inter-school championship, the Golden Brick) in Homeland Green; 'Register your school'.
- Schools and partners: four offers (free Engineering Day, after-school club, curricular program, fairs and the championship); 'Request a school proposal' mails admin@funademy.io. No prices, no partner names that are not signed.
- Contact: WhatsApp, email, social handle; 'Engineering and coding in person in Riyadh — SkyLab online for everyone'. No street address or phone list beyond the WhatsApp link.

6 · Component Library

مكتبة المكونات

Component	Anatomy	States and rules
Gold button	Pill, Baloo 800, gold fill, 5 px gold-shadow, shine sweep	Press: moves down 4 px; focus: cyan ring; one per viewport is primary — ghost (outlined) for the second action
Glass card	6% white fill, 14% white border, 10 px blur, 24 px radius	Night sections only; never on sand
Sand card	White, 1 px sand-grey border, 24 px radius	Sand sections only
Sand stripe	160 × 6 px gradient bar	Under every section title; the only divider on the site
Live ticker	Cyan pulse dot · LIVE · day time — program · [seats] in gold	From the admin schedule; hides when no slots are open; stops under reduced motion
Age rail	15 round buttons 4–18, gold when selected, horizontally scrollable	Default 8; updates the match panel without page reload; keyboard operable
Program tile	Navy icon square, field tag, name, meta line, one sentence	SkyLab tiles use the cyan tag
Altitude ladder	Five rungs rising left-to-right (right-to-left in Arabic), sand → gold	Ages and minutes on each rung
Phase strip	Four cells, sand ladder then Homeland Green	Campaign only
Quote card	Quote, who (first name + Riyadh), placeholder tag until real	Real people only in production; consent on file
FAQ item	Details/summary accordion, Baloo summary	First item open by default
Form field	12 px radius, sand-grey border, cyan focus	Errors state what to fix in the field's own words
Credit block	Fixed credit line, originality line, copyright	Text never rewritten; the only place the licensed mark appears

7 · Copy Deck

دليل النصوص

Where	Arabic	English
Hero	لا نعلّم أطفالنا عن الهندسة — بل نصنع منهم مهندسين.	We don't teach kids about engineering — we make them engineers.
Tagline	حيث تلتقي المتعة بالأكاديمية	Where fun meets academy
Primary CTA	احجز الحصة التجريبية المجانية	Book the free trial class
Header CTA	احجز حصة مجانية	Book a free class
Screens line	لا نستخدم الشاشات للاستهلاك، بل للابتكار	Screens for creation, not consumption
Campaign	أطفالنا يبنون وطننا	Our children build our nation
Age rail	كم عمر طفلك؟	How old is your child?
Price	1,380 ريال / طفل / برنامج شهريًا · الحصة الأولى مجانية	1,380 SAR / child / program a month · first class free
Contact	نخدم مدينة الرياض حضوريًا — وأونلاين للجميع.	We serve Riyadh in person — and everyone online.
Confirmation	تم استلام طلبك. سنتصل بك خلال 24 ساعة لتأكيد موعد الحصة المجانية.	Request received. We will call within 24 hours to confirm the free class.

Voice rules on the site

We say «نصنع مهندسين», «حصص مباشرة مع معلم حقيقي», «الحصة الأولى مجانية», «وقت شاشة يبني المهارات». We never say «دورات ترفيهية», «فيديوهات تعليمية», «خصومات حصرية لفترة محدودة», or anything that frightens parents about screens. Sentence case, plain verbs, no exclamation marks, no all-caps labels above headings.

8 · Conversion Flows, Technology and SEO

مسارات التحويل والتقنية وتحسين محركات البحث

Free trial flow

مسار الحصة التجريبية

- Visit → age rail or program page → 'Book the free class' → form (or WhatsApp) → record saved, admin notified → call within 24 hours → trial slot confirmed by WhatsApp with the parent card → trial → enrolment link (Moyasar/HyperPay: Mada, cards, STC Pay, Tamara) → ZATCA e-invoice → kit dispatched → first session.
- Every gold button and WhatsApp link fires an analytics event with the page and program; the admin dashboard shows bookings by source.

Technical specification

المواصفات التقنية

Layer	Choice
Framework	Next.js 14 App Router, TypeScript, server components; Arabic at / and English at /en through a locale middleware; the brand design system as one global stylesheet (globals.css) rather than a utility framework, so the tokens are the code
Data	PostgreSQL through Drizzle ORM inside the site (src/db/schema.ts): programs, slots, bookings, testimonials, faqs, events, settings — hosted on the same Saudi PostgreSQL as the Learning Hub, or handed to it later through the same API shapes
Hosting	Saudi-region hosting for PDPL residency; CDN for static assets; HTTPS; funademy.io with /en, no separate domain
Fonts	Baloo Bhaijaan 2 (700/800) and IBM Plex Sans Arabic (400–700) self-hosted, subset, preloaded; Arial fallback
Images	Real class photos within the first month (AI placeholders at launch, navy overlay 55–75% under text); WebP/AVIF, lazy-loaded; written parent consent before any child's photo
Forms	Server validation, rate limit, honeypot; notification to admin by WhatsApp Business API and email; no third-party form tools
Analytics	Privacy-respecting analytics (server-side events): trial_form_submit, whatsapp_click, age_rail_select, program_view, skylab_reserve, school_request
Payments	Handled on the platform after the trial (Moyasar/HyperPay) — the marketing site never takes card data
Admin	Password-protected /admin: slots CRUD (feeds the ticker and the booking form), bookings list with CSV export and contacted/enrolled flags, testimonials and FAQ editor, counters

SEO and metadata

تحسين محركات البحث

- Titles: 'FunAdemy — برامج هندسة ورياضيات وعلوم مباشرة للأعمار 4–18 | حيث تلتقي المتعة بالأكاديمية' (AR) and the English equivalent on /en; per-page titles and descriptions; hreflang ar/en; canonical per locale.
- Structured data: Organization (Falcon Eye Ltd, Riyadh), Course for each of the twelve programs, FAQPage on the home FAQ, Event for Mini Watan.
- Open Graph and Twitter cards with the navy kite image; a sitemap.xml and robots.txt; Arabic keyword targets around 'برمجة للأطفال الرياض', 'روبوتات للأطفال', 'حصص رياضيات أونلاين', 'STEM للأطفال'.

- Performance targets: Largest Contentful Paint under 2.5 s on 4G, Cumulative Layout Shift under 0.1, Lighthouse accessibility 95+.

Accessibility and compliance

إمكانية الوصول والامتثال

- Contrast: white on navy and Navy Ink on gold pass AA; body text never in gold, cyan or sand; visible cyan focus rings; the whole site works with a keyboard and a screen reader in both directions.
- prefers-reduced-motion stops the flip, the sway, the shine and the ticker.
- PDPL: a one-line notice under every form, data stored in the Kingdom, a privacy page, no third-party pixels without consent; ZATCA-compliant invoicing lives on the platform.

9 · The Coded Site: Backend, Data and Admin

الموقع المبرمج: الخلفية والبيانات ولوحة الإدارة

The design is now code: a Next.js application with its own PostgreSQL backend, delivered as the funademy-web repository. Every element of the prototype exists as a component reading from the database — the ticker reads open slots, the age rail and the booking form read the twelve programs, the program pages read their slots and related programs, testimonials appear only once approved, FAQs are edited in the admin. The build passed, the site was run against a local PostgreSQL, and two trial requests were submitted end to end — one through the API and one through the form — and appeared in the admin pipeline.

صار التصميم كودًا: تطبيق Next.js بخلفية PostgreSQL خاصة به، مسلَّم كمستودع funademy-web. كل عنصر في النموذج موجود كمكوّن يقرأ من قاعدة البيانات — الشريط المباشر يقرأ المواعيد المفتوحة، وسكة الأعمار ونموذج الحجز يقرآن البرامج الاثني عشر، وصفحات البرامج تقرأ مواعيدها والبرامج المجاورة، والشهادات لا تظهر إلا بعد اعتمادها، والأسئلة تُحرَّر من لوحة الإدارة. البناء نجح، وشُغِّل الموقع على PostgreSQL محلي، وأُرسل طلبان تجريبيان من البداية إلى النهاية وظهرا في لوحة الإدارة.

Architecture

البنية

Layer	Implementation
Routing	src/middleware.ts rewrites / to the Arabic segment and leaves /en; canonical Arabic URLs have no prefix; /ar redirects to /
Pages	src/app/[locale]/… — home, programs, programs/[slug], skylab, pricing, book, campaign, schools, contact, privacy, terms, admin, not-found; every page renders on demand from the database
Components	Header (logo flips to the cap on hover and keyboard focus), Footer (fixed credit line), Ticker, AgeRail, BookingForm, MobileCta (sticky book + WhatsApp bar under 960 px), Analytics (first-party beacon), KiteCap (official mark geometry)
API	src/app/api/* — public reads, the bookings and events writes, and the admin CRUD behind a signed HTTP-only cookie
Data	Drizzle ORM schema in src/db/schema.ts; SQL migrations in drizzle/; scripts/migrate.ts and scripts/seed.ts
Notifications	src/lib/notify.ts — email through SMTP and an optional WhatsApp Business webhook; logged when unconfigured
Security	zod validation on every write, per-IP rate limits, a honeypot on the form, hashed IPs only, HMAC-signed admin session, security headers, no third-party scripts

Data model

نموذج البيانات

Table	Columns (main)	Used by
programs	slug, name, track (engineering | skylab), field AR/EN, age_min, age_max, session_min, class_cap, format AR/EN, one_liner AR/EN, description AR/EN, builds JSON, sort_order, active	Age rail, tracks, program pages, booking form, sitemap
slots	program_slug, weekday, time, capacity, seats_left, format AR/EN, is_open, starts_on	Live ticker, program pages, admin

Table	Columns (main)	Used by
bookings	parent_name, phone, child_age, program_slug, preferred_time, city, notes, locale, source, status, admin_note, ip_hash, user_agent, created_at	Booking form → admin pipeline → CSV
testimonials	quote AR/EN, who AR/EN, approved, sort_order	Home (approved only)
faqs	q AR/EN, a AR/EN, page, sort_order, active	Home FAQ and FAQPage structured data
events	name, page, program_slug, locale, meta, created_at	Stats: page_view, book_cta_click, whatsapp_click, trial_form_submit
settings	key, value	Counters and flags

API reference

مرجع الواجهة البرمجية

Endpoint	Behaviour
GET /api/programs · /api/slots · /api/faqs?page= · /api/testimonials	Public JSON reads (testimonials: approved only)
POST /api/bookings	Validates name, phone pattern, age 4–18, known program; stores the request, logs a trial_form_submit event, notifies admin; 201 { ok, id }; 429 after five requests in ten minutes from one IP
POST /api/events	First-party analytics beacon from the browser; rate-limited
POST /api/admin/login · /logout	Password checked in constant time; 12-hour signed cookie
/api/admin/bookings (+?format=csv), /:id	Pipeline list, CSV export (UTF-8 with BOM for Excel), status and note updates
/api/admin/slots, /api/admin/testimonials, /api/admin/faqs (+/:id)	Create, update, delete; slots feed the ticker immediately
GET /api/admin/stats	Requests by status and program; events of the last 30 days

Admin dashboard

لوحة الإدارة

- Bookings: every trial request with its program, preferred time and notes; a status pipeline (new → contacted → trial booked → enrolled → closed) and an admin note per row; one-click CSV export.
- Slots: add, open/close, and adjust seats left — the live ticker and the program pages change on the next page load.
- Testimonials: added unapproved; only approved (real, consented) quotes reach the home page — the brand rule enforced by the data.
- FAQs: bilingual questions per page; stats: requests by status and program, and the first-party events of the last 30 days.

Run and deploy

التشغيل والنشر

- npm install · npm run db:migrate · npm run db:seed · npm run dev — or npm run build && npm start for production on port 3000 behind HTTPS.

- Environment: DATABASE_URL (Saudi-hosted PostgreSQL), ADMIN_PASSWORD, AUTH_SECRET, SITE_URL, optional SMTP_* and NOTIFY_EMAIL, optional WHATSAPP_WEBHOOK_URL, NEXT_PUBLIC_WHATSAPP.
- Replit: import into the existing FunAdemy project on the founder's go-ahead, attach PostgreSQL, add the secrets, migrate and seed, build and start — no new project is created.
- Fonts: self-host Baloo Bhaijaan 2 and IBM Plex Sans Arabic under public/fonts before launch for full data residency; the layout currently links Google Fonts.

10 · Launch Checklist and QA Gate

قائمة الإطلاق وبوابة الجودة

- Brand grep across the repository: zero district or neighbourhood names, zero 'territory' language, zero transliterated brand name, zero SAR 330/380 or per-track prices, zero multicolour stripes, the credit line verbatim.
- The licensed mark appears only in the footer credit line; SkyLab pages carry the originality line.
- All gold buttons lead to /book; WhatsApp links open the business number; the form saves, notifies and confirms in place; the ticker reads from live slots.
- Mobile at 360, 390 and 430 px; desktop at 1366 and 1920; RTL and LTR both checked on every page; Arabic line breaks reviewed on headings.
- Legal pages published; DNS and HTTPS on funademy.io; Instagram, YouTube and LinkedIn profiles link back; Google Business Profile for Riyadh.
- Testimonials: real names, real words, consent on file — or the section is hidden at launch.
- Lighthouse run and recorded; analytics events verified; admin login tested; backups scheduled.

Handover note

The prototype is the visual and copy source of truth; the developer rebuilds it in Next.js against the Learning Hub API without changing tokens, copy or the section order. Any Replit build is started only after the founder's go-ahead, by updating the existing project rather than creating a new one.

11 · Screens

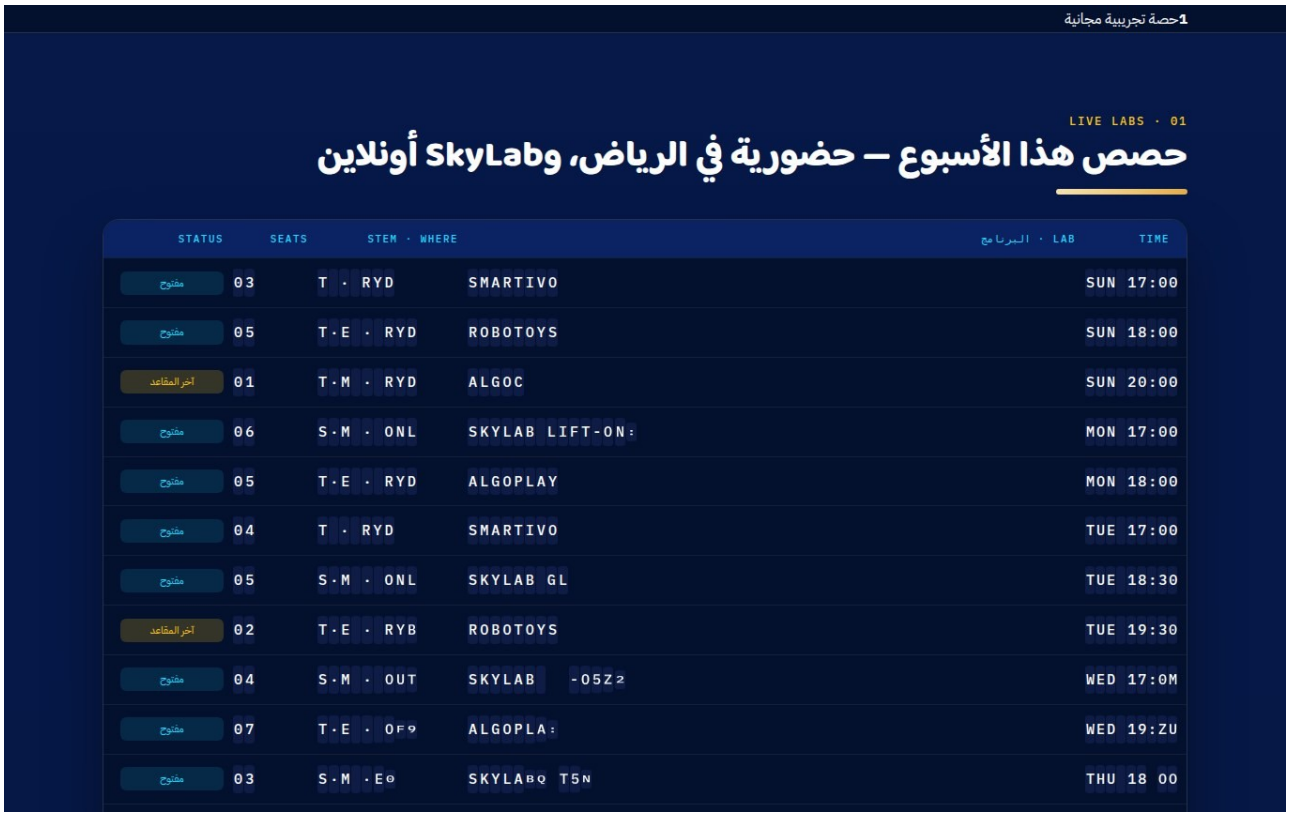
الشاشات



Hero — navy ground, gold second line, the kite flying across the top lane, the STEM playground



The header mark flips into the graduation cap under the mouse



Live labs board



STEM map — cyan · white · gold · sand lines, the dashed gold path to 18, paper STEM cards



Classroom band — brand-style scenes in the photo slots (to be replaced by real class photos)



Four pillars



The FunAdemy loop



One price ticket on paper with the gold stub



National campaign route on the sand ladder



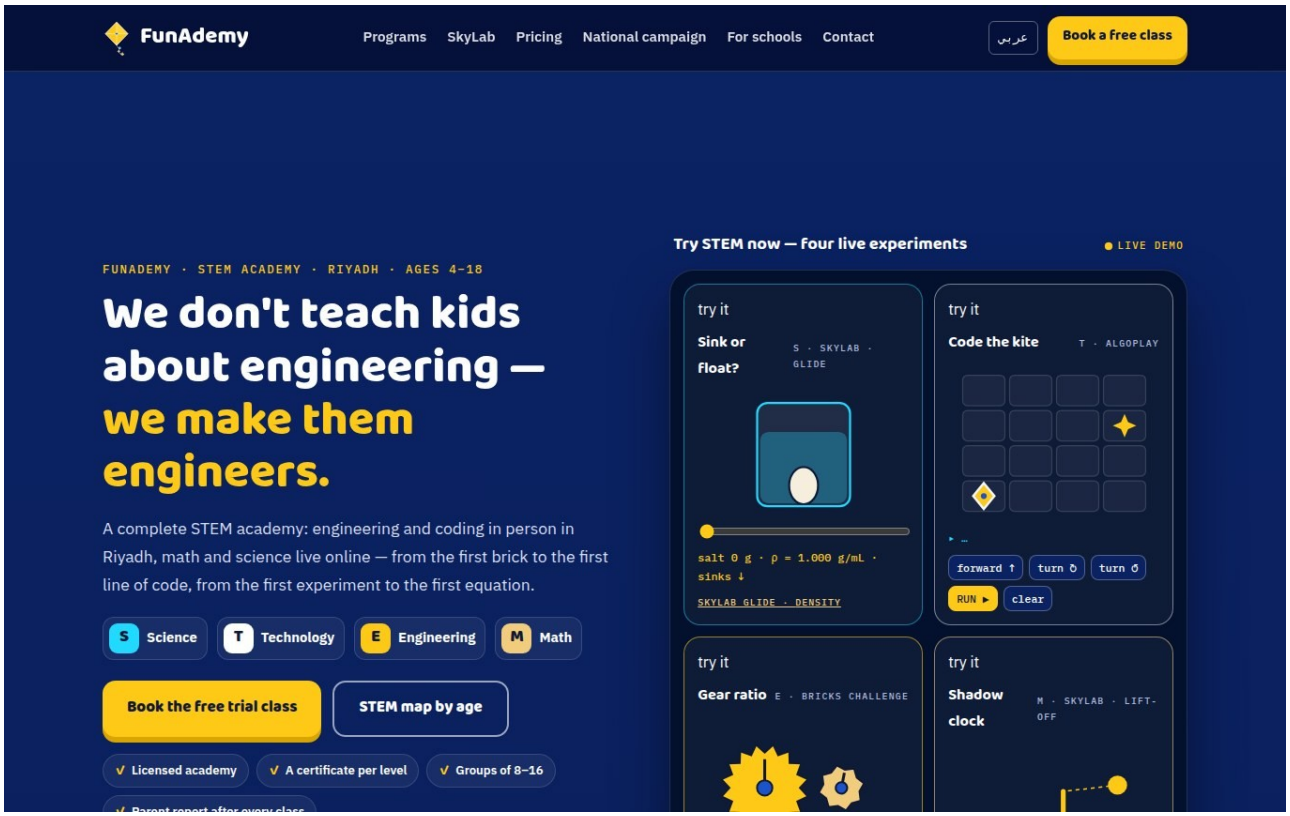
Program page



SkyLab — the fifty missions



Booking



English at /en



Mobile — hero, classroom band, STEM map

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Internal — founder review. The prototype and this specification are original FunAdemy work.